DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
MONDAY TEST

CLASS: X **FULL MARKS: 40**
SUBJECT: BIOLOGY **DATE: 14.7.22**

General Instructions:

- **The paper consists of three printed pages.**
- **Answers should be to the point.**
- **Copy the question number carefully before answering the questions.**
- **Marks will be deducted for spelling errors.**

1. Choose the correct answer: **[5]**

a. **The space between the cell wall and plasma membrane in a plasmolysed cell is filled with**
i. **Isotonic solution,** ii. **Hypotonic solution,** iii. **Hypertonic solution,** iv. **Water.**

b. **The rate of transpiration is less when**
i. **Humidity is high,** ii. **Atmosphere is dry,** iii. **Temperature is high,** iv. **Windy day.**

c. **The process by which water is lost from the margin of a leaf is known as**
i. **Guttation,** ii. **Transpiration,** iii. **Bleeding,** iv. **Absorption**

d. **Transpiration taking place through the minute openings on the surface of old stems**
i. **Cuticular Transpiration,** ii. **Lenticular Transpiration,** iii. **Stomatal Transpiration,** iv. **Stoma Transpiration**

e. **Sunken stomata is found in**
i. ***Hibiscus*,** ii. ***Nerium*,** iii. **Pine,** iv. **Bacteria**

2. Give reasons for the following: **[5]**

i. **Plasma membrane is known as selectively permeable membrane of a cell.**
ii. **Soaked seeds when placed in a fully closed container burst open with great pressure.**
iii. **The roots of certain trees have been seen to crack the walls.**
iv. **Desert plants have thick cuticle.**
v. **Transpiration is greater in a broad leaf than in a narrow leaf.**

3. **Differentiate between the following according to the criterion given in brackets:** **[5]**

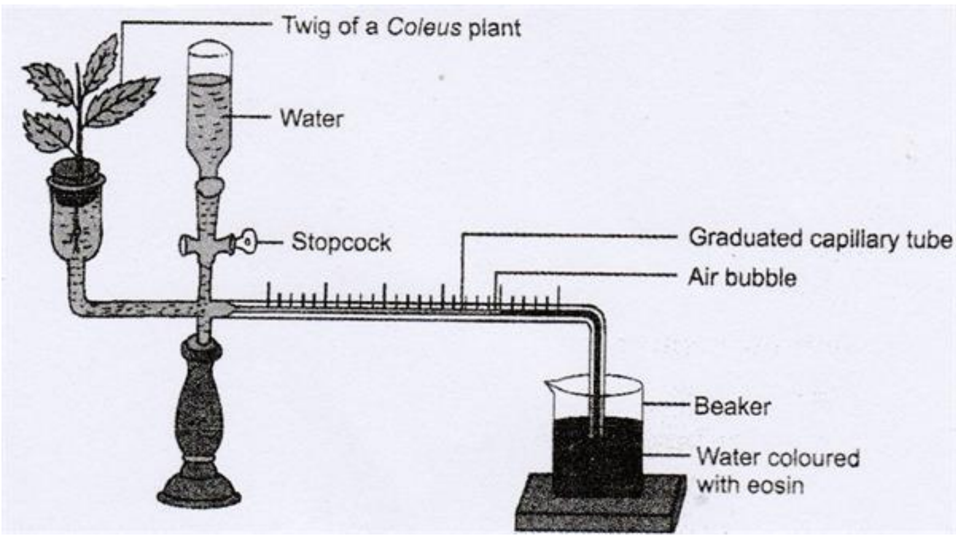
   i. **Evaporation and Transpiration (Definition)**
   ii. **Osmosis and Diffusion (Semipermeable membrane)**
   iii. **Active Transport and Passive Transport (Definition)**
   iv. **Plasmolysis and Deplasmolysis (Definition)**
   v. **Turgidity and Flaccidity (Definition)**

4. **Draw a neat and well labelled diagram of stomata. Mention how intensity of sunlight and atmospheric pressure affects transpiration.** **[5]**

5. **Name the following:** **[5]**

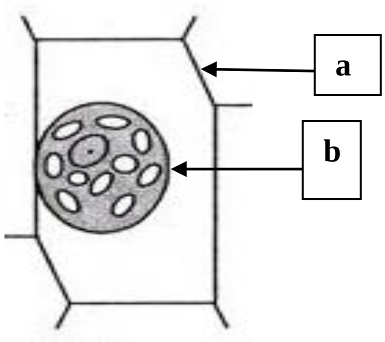
   i. **The non-living cell inclusion inside a cell which contains cell sap.**
   ii. **An instrument used to find the rate of transpiration.**
   iii. **The solution where the relative concentration of water molecules on either side of the cell membrane is same.**
   iv. **The vascular tissue which carries water and minerals from root to different parts of the plant.**
   v. **Absorption of water by the plant cells by surface attraction.**

6. **Observe the diagram given below and answer the following questions:** **[5]**



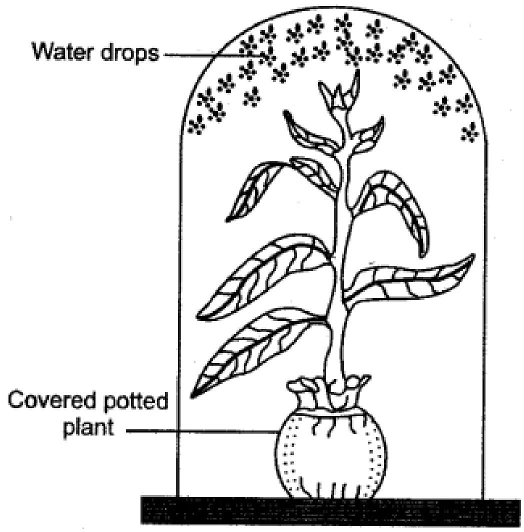
   i. **Identify the instrument.**
   ii. **Why do we use coloured water in this experiment?**
   iii. **Mention the utility of water reservoir in this experimental set up.**
   iv. **Mention two limitations of this instrument.**

7. Observe the diagram given below and answer the following questions: [5]



i. **Identify the state of the cell.**
ii. **Mention the tonicity of the liquid which is surrounding the cell.**
iii. **How will you change the state of the cell?**
iv. **Label the parts "a" and "b".**
v. **Mention one example of a semipermeable membrane.**

8. Observe the experiment given below and answer the following questions: [5]



i. **Mention the aim of the experiment.**
ii. **Why is the pot of the plant covered?**
iii. **How will you prove that the droplets of liquid inside the bell jar are water droplets only?**
iv. **Suggest a suitable control for this experiment.**
v. **Mention one significance of the process shown in this experiment.**